MILITARY SPECIFICATION
N-BENZYLDIMETHYLAMINE

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers one grade of benzyldimethylamine.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

(When requesting any of the above documents, give the title and complete designation of the item shown above. Copies of this specification and other unclassified specifications, standards and publications required by contractors in connection with specific procurement functions may be obtained from the Commanding Officer, Navy Supply Depot (CDS), 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

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PUBLICATIONS

United States Pharmacopoeia XVI

(Applications for copies should be addressed to the Mack Publishing Company, Easton, Pennsylvania.)

3. REQUIREMENTS

3.1 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the benzyldimethylamine shall be manufactured using the methods and procedures proposed for the production. The sample shall be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract. After satisfactorily passing all the preproduction tests specified herein, no changes in raw materials and processing of materials for production shall be made without the prior written approval of the procuring activity.

3.2 Data requirements. No data is required by this specification or by applicable documents referenced in Section 2 unless specified in the contract or order (see 6.2).

3.3 Material. The material shall be a colorless to light yellow liquid aromatic, tertiary amine.

3.4 Physical and chemical properties

3.4.1 Refractive index. The refractive index shall be no less than 1.4985 nor more than 1.5005 when measured at 25 ± 0.1 degrees Centigrade (C).

3.4.2 Assay. The N-benzyldimethylamine assay shall be no less than 97 percent.

3.4.3 Distillation range. The distillation range shall be 95 percent by weight from 177 to 181 degrees C.

3.4.4 Moisture. The moisture shall be no greater than 0.5 percent by weight.

3.5 Workmanship. The material shall be uniform in quality and shall be free from impurities and other defects that could adversely affect its use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the benzyldimethylamine shall be classified as follows:

a. Preproduction inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Sampling

4.3.1 Preproduction sample. A preproduction sample of sufficient quantity of the benzyldimethylamine manufactured in accordance with 3.1 shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Further production of the benzyldimethylamine by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Quality conformance inspection sampling. Unless otherwise specified, sufficient material shall be taken from each lot to perform the tests as specified in 4.5.

4.3.3 Inspection lot. An inspection lot of benzyldimethylamine shall be all the material offered for acceptance at one time and produced in a single manufacturing run under homogeneous conditions of manufacture.

4.4 Preproduction inspection. The preproduction sample shall satisfactorily pass the quality conformance inspections detailed in 4.5.

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4.5 Quality conformance inspection

4.5.1 Refractive index test. The refractive index shall be determined by use of an Abbe Refractometer.

4.5.2 Assay test. Weigh a 0.5-gram sample to the nearest 0.1 milligram. Place sample into a 250-milliliter (ml) Erlenmeyer flask. Add 10 ml of methanol (AR grade), 50 ml of distilled water, and titrate with 0.1 normal (N) sulfuric acid to the equivalence point using methyl purple as the indicator.

Calculations:

$$\text{Percent assay} = \frac{N\,(A) \times 0.135}{W} \times 100$$

where:

N = Normality of sulfuric acid used
A = Milliliters of sulfuric acid used
W = Weight of sample, grams
0.135 = Milliequivalent weight of N-benzyldimethylamine

4.5.3 Distillation range test. The distillation range shall be determined by method I of U. S. Pharmacopoeia XVI.

4.5.4 Moisture test. Weigh approximately 25 grams of material into a clean, dry titration flask on the Karl Fischer Titrimeter. Dissolve the sample in 20.0 ml of dry methanol measured by buret. Add excess Karl Fischer Reagent and back titrate with a water-methanol standard.

Calculations:

(ml KF Reagent) - (Methanol factor) - (ml water-methanol × R) = Net ml KF Reagent

$$\frac{\text{Net ml KF Reagent} \times \text{Fkf} \times 100}{\text{Sample wt. in grams}} = \text{percent } H_2O$$

Methanol factor = ml of KF Reagent to neutralize 20.0 ml dry methanol
R = ml of KF Reagent to neutralize 1.00 ml of water-methanol Std.
Fkf = Grams of H_2O each ml of KF Reagent will titrate

4.5.5 Visual examination. Visually examine all the containers in a lot to determine conformance to 3.5 and Section 5.

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4.6 Acceptance criteria

4.6.1 Preproduction. Failure of any sample to conform with any requirement of this specification shall be cause for rejection of the sample.

4.6.2 Quality conformance. Failure of any sample to meet any requirement of this specification shall be cause for rejection of the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Unless otherwise specified by the procuring activity, the preservation, packaging and packing shall be in accordance with MIL-P-116, method III.

5.2 Marking. Each container shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

a. Manufacturer's name and location
b. Material trade name
c. Net weight or volume
d. Lot number, batch number and date of manufacture
e. Shelf life or storage limitations
f. Number and revision letter of this specification

6. NOTES

6.1 Intended use. The material purchased in accordance with this specification is intended to be used as an accelerator in epoxy casting resins using anhydride curing agents.

6.2 Ordering data. Procurement documents should specify, but not be limited to, the following information:

a. Title, number and revision letter of this specification
b. Minimum lot size, if applicable
c. Whether preproduction sample is required
d. Place of delivery
e. Size of container
f. Request for test data

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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